

LECTURE ACTIVITY 1: STATISTICAL STORYTELLING

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Learning Objectives:

By the end of this activity, you should be able to

- 1) Define and identify observational units.
 - 2) Define and identify variables.
 - 3) Classify variables as categorical or quantitative.
 - 4) Extract basic information from simple data visualizations.
 - 5) Interact with and manipulate variables within plots.
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Welcome!

Welcome to our first Lecture Activity! To embody Cal Poly's spirit of "Learn by Doing," we'll frequently (almost every lecture) incorporate a Lecture Activity like this one that is designed to give you a more hands-on approach to the course material than a "standard" lecture would. I (Dr. Ethan) highly encourage you to work in groups, as statistics is meant to be a very collaborative field!

Part 1: A Video

Watch the video found at the [following link](https://bit.ly/1110-1a01) (for those of you who printed out this activity, here is the url: <https://bit.ly/1110-1a01>). We'll use the plot in this video to answer a few questions.



Question 1

What's the "story" this plot tells us? (This is an intentionally open-ended question!)

Let's break down this plot a bit more.



Question 2

What real-world object does each point (bubble) represent? **Hint:** what happens when you hover your cursor over each bubble?



Question 3

Why are some points bigger than others? Why are the sizes of the points changing as the video progresses?

Now, let's personalize a bit! That is, it's time for you to pick a country you're interested in to examine further.



Question 4

Select a country from the countries listed to the right side of the page and play the video again. Do you notice anything surprising in the country's trajectory regarding life expectancy? Can you justify any of these "surprises" based on country's history? Explain.

Statistical Terminology

Alright, let's start making things a little more formal. First, recall the following two pieces of vocabulary:

- **Observational Units** are the objects on which data is taken. These can be things like people, households, countries, cats - you name it!
- A **variable** is an attribute of an observational unit that we can measure. Note that the value of a variable can vary from unit to unit - for example, we expect different people to have different heights.



Question 5

Describe the observational unit (i.e. identify what the observational units are).



Question 6

What are the five variables represented in the plots?

Additionally, recall that variables come in (broadly speaking) two types:

- **Categorical variables** are variables whose values are “words-” more specifically, whose values fall into a fixed set of “categories” (which we often call **levels**).
- **Quantitative variables** are variables whose values are numbers, and those such that it makes interpretive sense to perform algebraic manipulations on the values.

⚠ Caution

Just because a variable has values that are numbers doesn't mean it's a quantitative one! For example, consider a dataset in which observational units are people and one variable is **Birth Month**. We often write birth months using numbers - e.g. “1” for January, “2” for February, etc. This doesn't make **Birth Month** a quantitative variable - that's because “1” + “2” really means “January” + “February,” which has no interpretive meaning.

In more technical statistical speak: categorical variable sometimes have values that are **encoded** as numbers.



Question 7

For each of the variables you identified in Question (6) above, classify them as either categorical or quantitative.

Part 2: Monetary Allocation of the Brazilian Government

We'll now take a look at some information published by the Brazilian government on expenditure allocation; specifically, we'll be focusing on a plot posted to their website that displays the five sub-categories within Hospital and Outpatient Care sub-area that received the most resources (presumably money) in 2021. A quick disclaimer: the original plots have text in Brazilian Portuguese, but we will be exploring the plots as translated to English by Google Chrome's auto-translation feature.

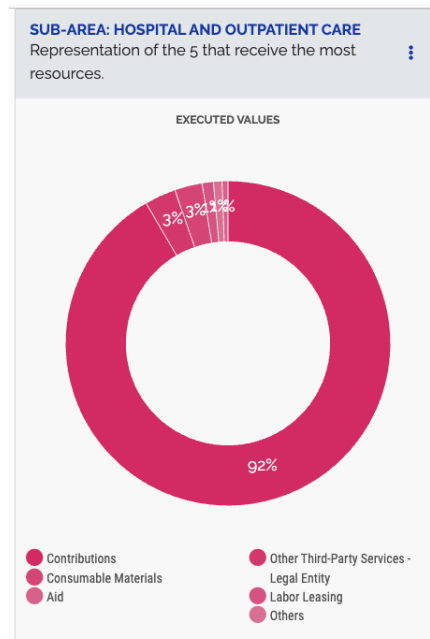


Figure 1: A donut chart of allocation percentages to the hospital and outpatient care sub-area of the Brazilian Government, in 2021. Image source: <https://portal.datransparencia.gov.br/funcoes/10-saude?ano=2021>



Question 8

What's the "story" this plot tells us? (This is an intentionally open-ended question!)

Another valuable skill to have is the ability to make constructive criticism of plots!



Question 9

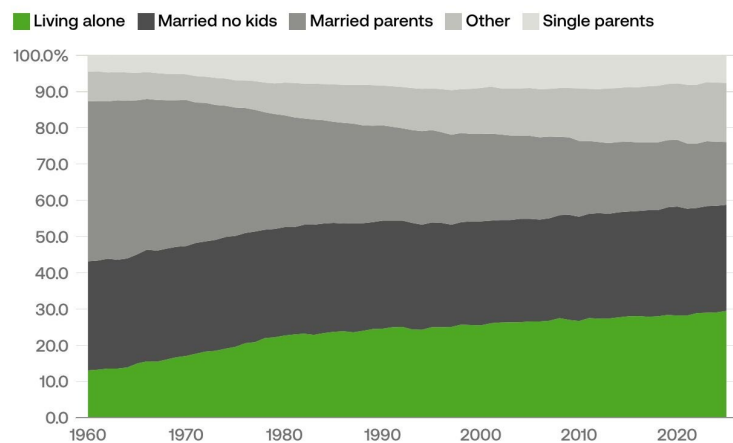
Was Figure (1) easy to read? If not, what specific aspects were difficult to read? How did those aspects impact your ability to identify the “story” the plot is trying to tell?

Part 3: US Living Situations

Below is a figure demonstrating how the different living situations of US households (living alone, Married with no Kids, etc.) has changed between 1980 and 2025.

Solo living topped all household types for the first time in 2025.

Share of households by household type, 1960–2025



"Parent" categories include parents living with their own children under the age of 18. "Other" includes family households (such as adult relatives) and nonfamily households (such as nonmarried partners or roommates).

Source: Census Bureau



Figure 2: A ribbon plot demonstrating how US households' living situations have changed over time. Categories include: Living Alone, Married No Kids, Married Parents, Other, and Single Parents. Image Source: <https://usafacts.org/articles/semiquincentennial-snapshot-the-us-at-250-years/>



Question 10

Describe the observational unit.



Question 11

What type of variable (categorical or quantitative) is **Household Type**? If it is categorical, what are its levels?

Let's practice extracting some more detailed information from plots.



Question 12

Approximately what percentage of households in 1960 were in the **Living Alone** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?



Question 13

Approximately what percentage of households in 2020 were in the **Living Alone** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?



Question 14

Approximately what percentage of households in 1960 were in the **Married Parents** category? How can you tell (i.e. what parts of the plot did you use to come to your answer)?